

Permutation Reconstruction Using Annealed Compatibility of Independent Hypotheses

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ABSTRACT

This note introduces a preliminary computational framework for reconstructing a permutation matrix from partially observed row and column statistics. The hidden object is an $(n \times n)$ matrix containing each integer from (1) through (n^2) exactly once. For every row and column, two statistics are defined: the sum of its' entries and the sum of the logarithms of its' entries. Only a random subset of these statistics is observed. The reconstruction problem is reformulated away from regression and towards a direct search approach. The initial model, called **Simple Swap**, maintains two independently initialized permutation hypotheses. One hypothesis attempts to satisfy the observed row statistics, while the other attempts to satisfy the observed column statistics. Each hypothesis is modified separately through simple pairwise swaps. The central mechanism is a simulated-annealing procedure that introduces time-dependent compatibility between the two hypotheses. At each iteration, the relative row and column reconstruction errors influence which hypothesis receives the next proposed move. The intention is to allow each marginal representation to make progress independently while progressively encouraging the two representations to become compatible descriptions of the same permutation. A Graph Neural Network (GNN) is used to propose swaps and GNN implementation is done using PyTorch. The initial implementation fixes $(n = 4)$, giving a permutation of $(1, \dots, 16)$. This deliberately simple setting is intended to test the underlying search principle before more sophisticated move operators, compatibility metrics, and learned search mechanisms are introduced.

Keywords: Combinatorial Search, Reinforcement Learning, Computational Reasoning

RESEARCH QUESTION

Can independently optimized row and column hypotheses, coupled through a time-dependent compatibility mechanism, provide a useful search strategy for exact permutation reconstruction?

PROBLEM

Consider an $(n \times n)$ matrix containing every integer from (1) to (n^2) exactly once. Rather than observing the matrix directly, we observe permutation-invariant quantities derived from its rows and columns. For each row and column, the observed quantities consist of the sum of its entries and the sum of the logarithms of its entries.

Only a random subset of these statistics are observed with independent and identical observation probability of (0.5) . The reconstruction problem is to find a permutation consistent with the available observations. The problem is therefore combinatorial in nature. The number of possible permutations grows factorially, while partial observations sets may correspond to multiple valid reconstructions.

CANONICAL FAMILY

The search space contains substantial symmetry because row permutations, column permutations, and transposition can produce equivalent representations of the same underlying structure. A canonical family is therefore introduced by selecting the lexicographically minimal representation under these transformations. The canonical representation reduces redundant exploration of equivalent states and provides a common representation for evaluating reconstructed solutions.

EXAMPLE

For the initial implementation, $n = 4$, so every hidden matrix is a reshaping of a permutation of $(1, \dots, 16)$. An instance is generated by sampling a canonical permutation matrix and then independently revealing each row and column sum and log-sum with equal and independent probability of (0.5) . The resulting observation pattern defines the search problem. The hidden matrix remains available during simulation so that the search procedure can be evaluated against the known ground truth.

APPROACH

The central idea is to replace a single global reconstruction process with two independent initialized hypotheses. The first hypothesis is initialized as a canonical permutation of $(1, \dots, 16)$ and optimized against the observed row statistics whereas the second hypothesis is initialized as a canonical permutation of $(1, \dots, 16)$ and optimized against the observed column statistics. Each hypothesis is then separately modified through simple swap operation in which two matrix entries are exchanged. The swap action preserves the permutation constraint because the state prior to swap is a valid permutation and the swap action is permutation invariant.

Hypothesis Selection

At each iteration, the relative row and column errors determine which hypothesis receives the next proposed action. Under this proposed formulation, the hypothesis with the greater current error receives more opportunities to change. This proposal rule is intended as a simple starting point rather than a fixed design choice.

Simulated Annealing

A simulated-annealing procedure controls acceptance of proposed actions. For a change in objective value (ΔE), the standard acceptance rule is

$$\begin{cases} 1, & \Delta E \leq 0, \\ \exp(-\Delta E/T_t), & \Delta E > 0, \end{cases}$$

where (T_t) is the temperature at iteration (t) . The important extension is that the annealing process is also used to introduce a progressively stronger compatibility relationship between the two hypotheses.

COMPATIBILITY

The two hypotheses are initially independent representations of different marginal views of the hidden permutation. However, a valid reconstruction must ultimately satisfy both row and column constraints simultaneously. This motivates a time-dependent compatibility mechanism. Conceptually, the combined search objective can be represented as:

$$E_r(X^{(r)}) + E_c(X^{(c)}) + \lambda_t K(X^{(r)}, X^{(c)})$$

where E_r measures row error, E_c measures column error, K measures incompatibility between two hypotheses and λ_t controls the strength of interaction over time.

Early in the search, the compatibility contribution is intended to be relatively weak, allowing the two hypotheses to explore their respective objective landscapes. As the search progresses, compatibility becomes increasingly important, encouraging the two hypotheses to converge toward a common permutation.

The precise form of (K) is an open research question. Potential formulations include indicator-positional disagreement, shared permutation structure, or compatibility measures derived from decomposed reconstruction errors.

LEARNED SWAP PROPOSALS

The Simple Swap framework also introduces a Graph Neural Network as a learned mechanism for proposing swaps. The GNN is implemented using PyTorch. The purpose of using a GNN is not initially

to replace the exact reconstruction objective. Instead, it provides a mechanism for learning which local permutation modifications may be useful given the current search state. The resulting architecture separates three components:

observed statistics → hypothesis state → learned swap proposal → annealed search

This provides a potential bridge between learned search heuristics and exact combinatorial reconstruction from partial signals. The GNN component is intentionally introduced without Monte Carlo Tree Search so that future improvements can be evaluated against a clearly defined baseline.

INITIAL RESEARCH DIRECTION

The initial implementation is designed to test whether the decomposition of, and interaction between, independently optimized row and column hypotheses provide a useful search signal.

The first comparisons should include:

1. Random permutation search.
2. Single-hypothesis swap search.
3. Independent row and column searches without compatibility.
4. Coupled row and column search with annealed compatibility.
5. Coupled search with learned GNN-based swap proposals.

The primary measurements should include exact reconstruction success, observed-statistic satisfaction, best reconstruction error, time to solution, and search trajectory. The trajectories of row error, column error, compatibility, temperature, and hypothesis-selection probability are particularly important because the objective is to understand the dynamics of the search mechanism rather than simply measure final accuracy.

WHY THIS MATTERS

The earlier formulation treated learned models primarily as predictors of properties of a combinatorial search problem. This project moves the learned component closer to the search process itself.

The broader computational framework can be represented as:

partial observations → hypotheses → learned proposals → search decisions → verified reconstruction

This creates a setting in which learning can influence where the search looks while the underlying permutation constraints provide an exact and independently verifiable objective.

The central question is therefore not simply whether a model can predict a useful quantity, but whether learned signals and interacting hypotheses can help navigate a difficult combinatorial search space.

FOLLOW-UP QUESTIONS

1. Can coupled row and column search outperform independent hypothesis search?
2. Does hypothesis crossover improve performance over simply coupled compatibility search?
3. Does error-weighted hypothesis selection improve search efficiency relative to uniform selection?
4. Can a GNN learn useful swap proposals from the current permutation and observed invariant structure?
5. Can learned swap proposals reduce the number of iterations required to find an exact reconstruction?
6. Can more sophisticated constructions of compatibility improve reconstruction performance?

7. Does the search mechanism transfer to larger permutation matrices?
8. Can the same framework be extended to additional combinatorial reconstruction problems?
9. Can reinforcement learning eventually learn useful search operators on top of the simple-swap framework?

KEY IDEA

The broader motivation is to study whether multiple incomplete computational hypotheses can be allowed to explore independently and then progressively coupled into a globally consistent solution.

The proposed mechanism is to propagate independent hypotheses that learn local proposals, which are then annealed together using compatibility to form a globally consistent reconstruction.

Permutation reconstruction provides a controlled environment for studying this idea because the underlying state is combinatorial, the observations are inexpensive to generate, and candidate solutions can be exactly verified. The Simple Swap model is therefore intended as a minimal experimental foundation rather than a finished algorithm.

Status: Computational experiment / preliminary research direction

Code: https://github.com/algorithmic-andy/permutation_reconstruction_simple_swap

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